

Service Publication

Diesel Engine
12V 396 TE74

Description and
Operation Manual
M011140/00E

gültig ab
valid from 1993

DETROIT DIESEL



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Notes for the user

This Manual consists of 10 Sections, marked by tab-index sheets.

These Sections are listed on this page.

Each Section has its own detailed Table of Contents which lists the Groups and Subgroups.

The Groups in Sections B and G correspond to the Groups in the Spare Parts Catalog and other Product Support publications.

Using the Table of Contents to trace an entry

- ➡ Table of Contents (general)
- ➡ Table of Contents (section)
- ➡ Group / Subgroup
- ➡ Heading

Using the Alphabetical Index to trace an entry

- ➡ Term in Alphabetical Index
- ➡ Group / Subgroup
- ➡ Heading

Publication Amendment Service

Please complete and return the Confirmation of Receipt card; you will then receive amendments to your manual.

A Product Summary
Engine summary, technical data

B Description
System and subassembly descriptions

C Operating Instructions
Task descriptions

D Maintenance Schedule
Schedule and maintenance frequency chart

E Troubleshooting
Fault conditions, probable causes, remedies

F Drawings
Engine cross-sections

G Task Description
Inspection and maintenance

K Preservation
Engine preservation

Z Appendix
Supplementary documentation

? Index
Alphabetical index of key terms

Amendment Service!

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M011140/00E

**Description and
Operation Manual**

12V 396 TE 74

Postcard

MTU Friedrichshafen GmbH
Department SCT
88040 Friedrichshafen
GERMANY

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Diesel Engine

Part A
Product Summary

The product summary gives details of the engine and contains technical data such as engine power, general engine specifications, engine weight, oil and coolant capacities, settings and operational data.

ENGINE LAYOUT, ENGINE POWERKey to Engine Model Designation 12 V 396 TE 74

12	=	number of cylinders
V	=	Vee configuration
396	=	engine series - 100 times displacement of one cylinder in litres
T	=	exhaust turbocharging
E	=	charge air cooling in split-circuit coolant system
7	=	marine engine
4	=	design index

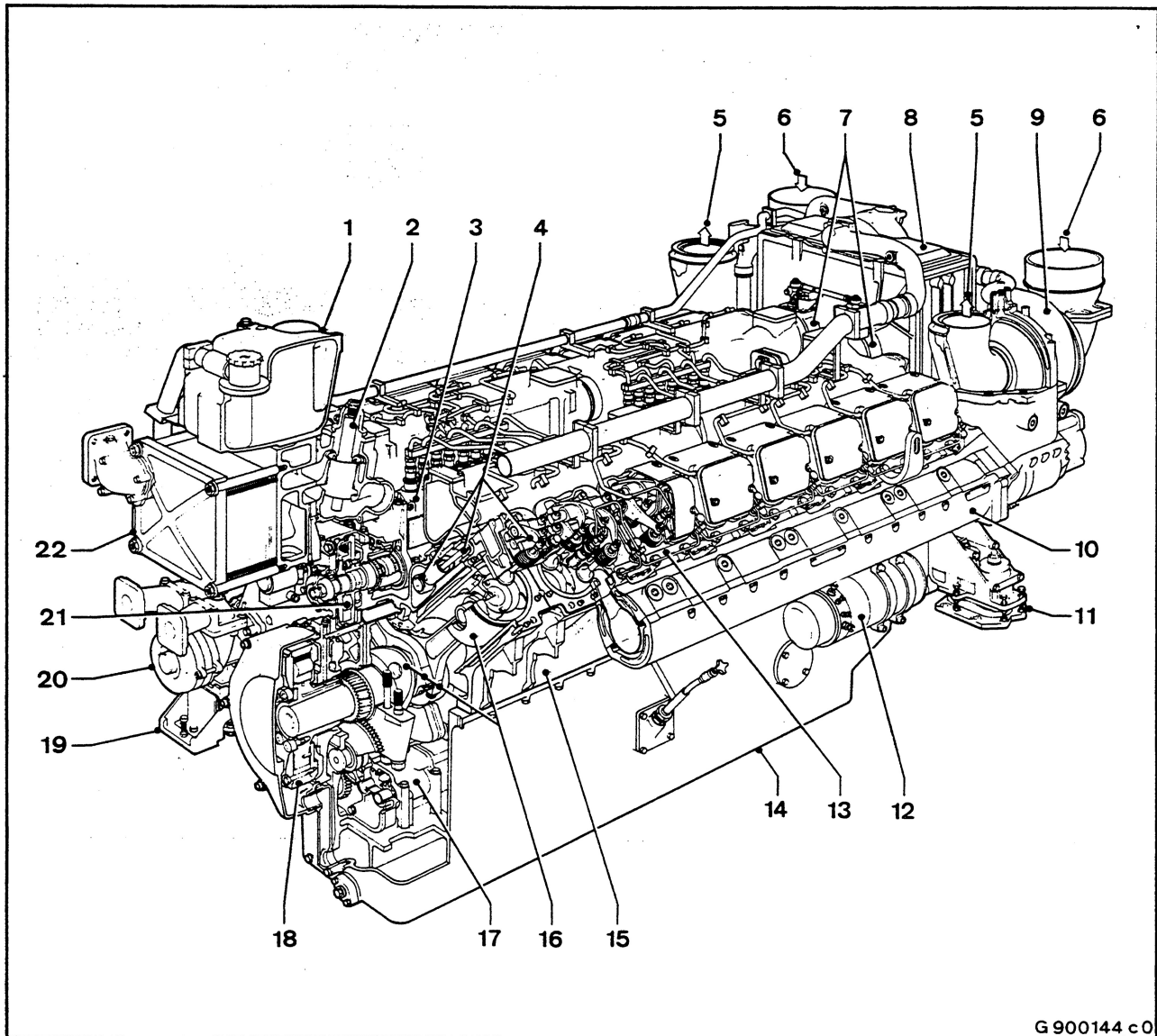
Engine power

Fuel stop power 1260 kW at 1900 rpm

Reference conditions

- Intake air temperature	25 °C
- Raw water temperature	25 °C
- Intake air pressure differential	20 mbar
- Exhaust pressure differential	30 mbar
- Barometric pressure	1000 mbar

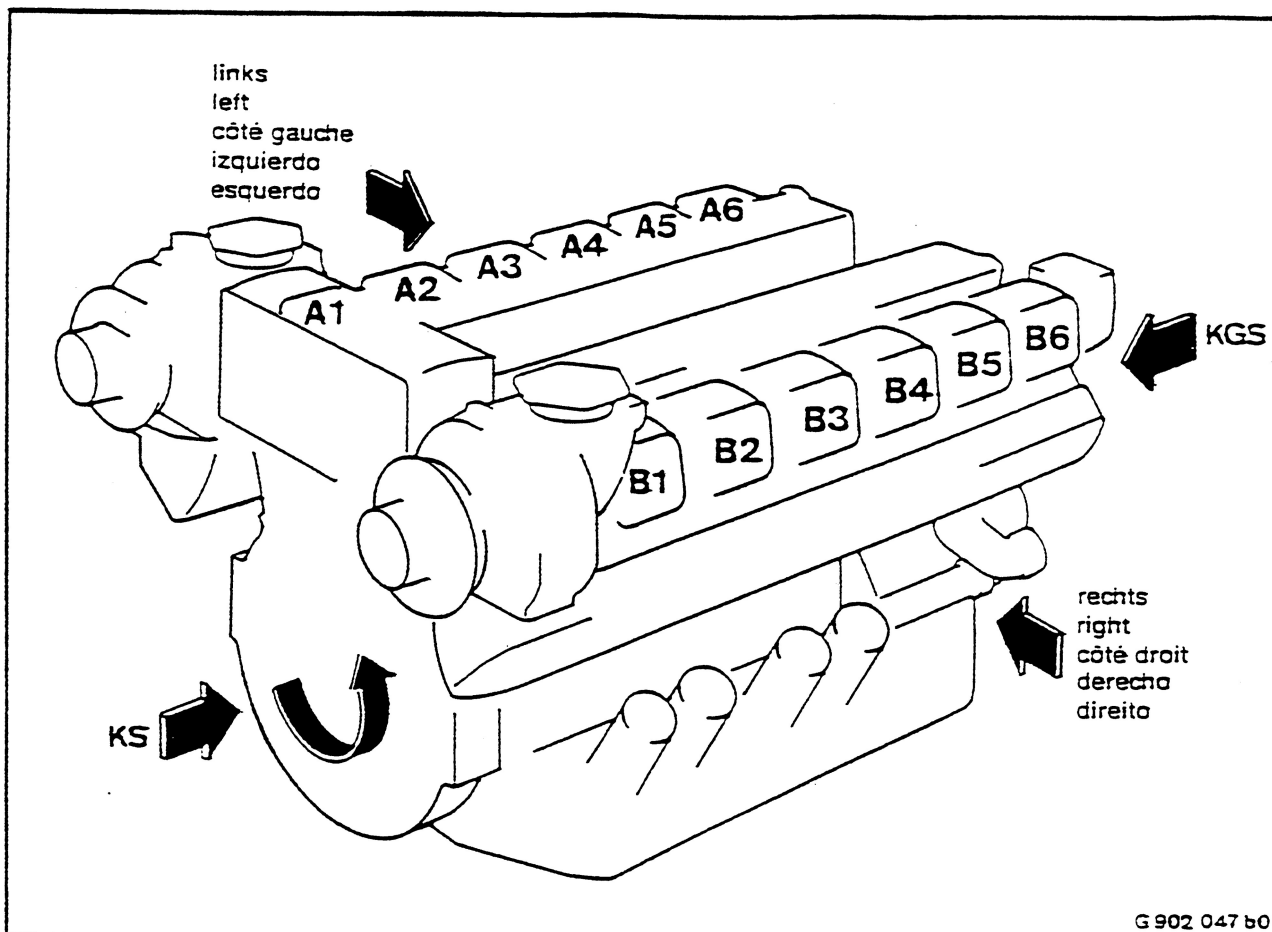
Engine Layout Cutaway Drawing 16V 396 TE.4 (also applies to 8 and 12V)



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- | | |
|-------------------------------|-----------------------------|
| 1 Coolant expansion tank | 12 Starter |
| 2 Coolant thermostat | 13 Cylinder head |
| 3 Fuel injection pump | 14 Oil pan |
| 4 Valve gear | 15 Crankcase |
| 5 Exhaust outlet | 16 Running gear |
| 6 Air inlet | 17 Engine oil pump |
| 7 Emergency air shut-off flap | 18 Vibration damper |
| 8 Intercooler | 19 Engine mount, timing end |
| 9 Exhaust turbocharger | 20 Bilge pump |
| 10 Exhaust manifold | 21 Fuel injection timer |
| 11 Engine mount, flywheel end | 22 Coolant cooler |

ENGINE SIDE AND CYLINDER DESIGNATIONS



Engine sides are always designated as viewed from the main PTO end.

The left bank cylinders are designated "A" and the right bank "B" (according to DIN/ISO Standard No. 1204).

The cylinders of each bank are numbered consecutively, starting with No. 1 at the main PTO end.

Numbering of other engine components is also from the main PTO end, starting with No. 1.

Designations and Abbreviations

Main PTO end	= KS (formerly HKS)
Auxiliary PTO end	= KGS (formerly GKS)
Left side	
Right side	
Top	
Bottom	

GENERAL SPECIFICATIONS

Operating method	Four-stroke cycle single-acting
Combustion method	direct injection
Mode of supercharging	Exhaust turbocharging Single-stage sequential turbocharging
Cooling method	liquid-cooled
Cylinder arrangement	90° Vee
Bore	165 mm
Piston stroke	185 mm
Displacement, cylinder	3.96 l
Number of cylinders	12
Displacement, total	47.5 l
Compression ratio	12.3 : 1
Direction of rotation, viewed from driving end	c.c.w. (non reversible)
Firing order	A1 - B2 - A5 - B4 - A3 - B1 - A6 - B5 - A2 - B3 - A4 - B6
Firing order, cylinder cutout	A1 - A5 - A3 - A6 - A2 - A4
Fuel injection pressure:	
Setting pressure with new spring	280 bar to 288 bar
Specified pressure with run-in spring	270 bar to 278 bar
Minimum pressure with run-in spring	260 bar
Final compression pressure at 120 rpm, 1000 mbar and 5 °C engine coolant temperature	20 bar to 24 bar
Breakaway torque at 40 °C engine coolant temperature, allowance for acceleration included	approx. 600 Nm
Cranking torque at 40 °C engine coolant temperature	approx. 560 Nm
Firing speed at 40 °C engine coolant temperature	approx. 100 to 120 rpm
Mean piston speed	11.7 m/s at 1900 rpm
Engine noise, mean free-field level	106 dB(A) 1 m distance
Exhaust noise, undamped free-field level	117 dB(A) 1 m from outlet